

A recursive dimension formula for $B_{\ell+1}^{(\lambda)} V_\lambda$ at multi-corner partitions

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Abstract

Let $H_q(S_n)$ be the Iwahori–Hecke algebra and V_λ its Specht module in the Hoefsmit seminormal basis. Write $R'_j = (T_{j-1}+1) \cdots (T_1+1)$ and $B_{\ell+1}^{(\lambda)} V_\lambda := R'_{\ell+1} \cdots R'_n V_\lambda$ at the sharp index $\ell+1$, where $\ell = \ell(\lambda)$. We propose and test a recursive formula expressing $\dim B_{\ell+1}^{(\lambda)} V_\lambda$ as a sum over length-preserving corners c_i of λ :

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \sum_{i=1}^r \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}, \quad \mu_i := \lambda \setminus \{c_i, c_*(\lambda)\},$$

where $c_*(\lambda) = (\ell, 1)$ is the column-1 bottom and r is the number of length-preserving corners. The base case ($r = 0$, i.e. a hook or near-hook) is $\dim = 1$. Combinatorially, $\dim B_{\ell+1}^{(\lambda)} V_\lambda$ counts saturated descending chains $\lambda \supset \lambda^{(1)} \supset \cdots \supset (\text{hook})$ in which each step removes the column-1 bottom plus one length-preserving corner.

We verify the formula computationally on nine families covering $r \in \{1, 2\}$ at $n \in \{8, \dots, 11\}$, all matching. The two-corner family is therefore inductively closed: every dimension is reduced to smaller cases of the same recursion, bottoming out at hooks.

Closed form. The recursion admits a clean closed-form solution via a bijection between saturated descending chains and standard Young tableaux of the *decoration shape* $\lambda^{(\geq 2)} := (\lambda_1 - 1, \lambda_2 - 1, \dots)$ (the partition obtained from λ by deleting column 1). The main result is

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda^{(\geq 2)}}$$

where f^μ is the number of SYT of shape μ . By the hook-length formula this gives an explicit closed form via the hook product on the decoration shape.

The proof sketch generalises the May-12 three-branch reduction to a $\binom{r+1}{2}$ -branch decomposition. Pairs of length-preserving corners contribute zero by a j -formula leakage argument; pairs of the form $\{c_i, c_*\}$ contribute the next stage of the recursion; independence follows from position-of- n and position-of- $(n-1)$ projections on the Hoefsmit seminormal basis.

1 Setup

1.1 Notation

Let q be generic. The *Iwahori–Hecke algebra* $H_q(S_n)$ has generators $T_1, \dots, T_{n-1}$ subject to $(T_i - q)(T_i + 1) = 0$ and the braid relations $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$, $T_i T_j = T_j T_i$ for $|i - j| \geq 2$. For each $\lambda \vdash n$, write V_λ for the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$. The q - and (-1) -eigenspaces of T_j are denoted E_j^+ and E_j^- .

Write

$$R'_j := (T_{j-1}+1)(T_{j-2}+1)\cdots(T_1+1) \in H_q(S_j),$$

and, for $\lambda \vdash n$ with $\ell := \ell(\lambda)$,

$$B_j^{(\lambda)} V_\lambda := R'_j R'_{j+1} \cdots R'_n V_\lambda.$$

We focus on the *sharp* index $j = \ell + 1$:

$$B_{\ell+1}^{(\lambda)} V_\lambda = R'_{\ell+1} R'_{\ell+2} \cdots R'_n V_\lambda.$$

1.2 Corners

A cell $c = (r, \lambda_r)$ is a *removable corner* of λ if $\lambda \setminus \{c\}$ is still a partition. We single out two types.

Definition 1 (Length-preserving corner). A removable corner $c = (r, \lambda_r)$ is *length-preserving* if $r < \ell(\lambda)$ and $\lambda_{r+1} < \lambda_r$. Equivalently, removing c leaves $\ell(\lambda)$ unchanged.

Definition 2 (Column-1 bottom). The *column-1 bottom* of λ is $c_*(\lambda) := (\ell(\lambda), 1)$. It is always removable (when $\lambda \neq (1^n)$); removing it decreases ℓ by 1.

If λ has r length-preserving corners $c_1, \dots, c_r$, then it has exactly $r + 1$ removable corners in total: the c_i together with c_* . (For hooks, $r = 0$; for staircases-like shapes, r can be large.)

1.3 Background

The starting point is the *rank-zero theorem* (Clio, 2026-05-08–10):

$$\Pi^{S_n}|_{V_\lambda} = 0 \quad \text{whenever } \ell(\lambda) > \lceil n/2 \rceil,$$

where $\Pi^{S_n} = B_2^{(\lambda)}$. Subsequent work (the *j-formula refinement*, Clio 2026-05-11) showed that at the sharp index $\ell + 1$,

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda},$$

and is genuinely the bottom non-vanishing stage of the iterated product. The May-11 *rank-corner formula* conjectured

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda \stackrel{?}{=} \#\{\text{length-pres corners of } \lambda\} = r.$$

This was refuted by Clio (2026-05-12, evening 3, paper 2026-05-12-evening-4-2-1n-dim3.tex) at $\lambda = (4, 2, 1^{n-6})$: the prediction is $r = 2$ but the actual dimension is 3. The proof there established a three-branch reduction yielding $\dim = \dim B^{(\mu_A)} + \dim B^{(\mu_B)} = 2 + 1 = 3$, where μ_A, μ_B are obtained by removing c_* together with the two length-preserving corners. This suggested a *recursive* replacement for the rank-corner formula, which we develop here.

2 The refined formula

Conjecture 3 (Refined dimension formula). *Let $\lambda \vdash n$ be rank-zero with $\ell = \ell(\lambda)$ and length-preserving corners $c_1, \dots, c_r$ ($r \geq 0$), and column-1 bottom $c_*(\lambda)$. Then*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \sum_{i=1}^r \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}, \quad \mu_i := \lambda \setminus \{c_i, c_*(\lambda)\}. \quad (1)$$

The base case ($r = 0$, λ a hook) is $\dim = 1$.

Note that μ_i is a partition of $n - 2$ with $\ell(\mu_i) = \ell - 1$, so the index $\ell - 1$ on the right is the *sharp* index $\ell(\mu_i) + 1 - 1 = \ell(\mu_i)$ shifted in the expected way: the iteration $B_{\ell-1}^{(\mu_i)}$ on V_{μ_i} (which lives at $\ell(\mu_i) + 1 = \ell$) is the sharp stage for μ_i . The recursion therefore closes within the family of rank-zero shapes.

Remark 4 (Combinatorial interpretation). Iterating (1) expresses $\dim B_{\ell+1}^{(\lambda)} V_\lambda$ as the number of *saturated descending chains*

$$\lambda = \lambda^{(0)} \supset \lambda^{(1)} \supset \dots \supset \lambda^{(k)} = (\text{hook}),$$

where each step removes the column-1 bottom of $\lambda^{(t)}$ together with one of its length-preserving corners:

$$\lambda^{(t+1)} = \lambda^{(t)} \setminus \{c_*(\lambda^{(t)}), c_{j_t}^{(t)}\}.$$

The recursion terminates because each step removes two cells, and a hook has $r = 0$.

Example 5 (Hooks and two-column shapes). If $\lambda = (k, 1^{n-k})$ with $k \geq 2$, then λ has $r = 1$ length-preserving corner (the $(1, k)$ cell when $k \geq 2$), and $\mu_1 = \lambda \setminus \{(1, k), (n - k + 1, 1)\} = (k - 1, 1^{n-k-1})$ is again a hook. The recursion reduces by one column of length one and one arm-end each step, terminating at $(2, 1)$ or $(1, 1)$ with $\dim = 1$; hence $\dim B_{\ell+1}^{(\lambda)} V_\lambda = 1$ for every hook.

Similarly, for $\lambda = (2^a, 1^{n-2a})$ with $a \geq 1$, there is a single length-preserving corner (the $(a, 2)$ cell) and the recursion descends through the family, again giving $\dim = 1$.

Example 6 (The $(3, 2, 1^{n-5})$ family: $r = 2$). For $\lambda = (3, 2, 1^{n-5})$, the two length-preserving corners are $c_1 = (1, 3)$ and $c_2 = (2, 2)$, and $c_* = (n - 4, 1)$. The two partners are

$$\mu_1 = (2, 2, 1^{n-6}), \quad \mu_2 = (3, 1^{n-5}).$$

Both are hooks (or hook-like), each contributing $\dim = 1$. Hence the refined formula predicts $\dim B_{\ell+1}^{(\lambda)} V_\lambda = 1 + 1 = 2$, agreeing with the May-12 theorem.

Example 7 (The $(4, 2, 1^{n-6})$ family: $r = 2$, asymmetric). For $\lambda = (4, 2, 1^{n-6})$, corners are $c_1 = (1, 4)$, $c_2 = (2, 2)$, $c_* = (n - 4, 1)$. The two partners are

$$\mu_1 = (3, 2, 1^{n-7}), \quad \mu_2 = (4, 1^{n-6}).$$

μ_1 is the $(3, 2, 1^*)$ family from Example 6, with $\dim = 2$; μ_2 is a hook, with $\dim = 1$. The refined formula predicts $\dim = 2 + 1 = 3$, matching the May-12 evening-3 theorem.

3 A closed-form formula via SYT count

The recursion of Conjecture 3 produces, after iteration, the number of saturated descending chains (Remark 4). We now identify this count with a familiar combinatorial object: the number of standard Young tableaux of a certain shape.

3.1 The decoration shape

Definition 8 (Decoration shape). For a partition $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$, the *decoration shape* $\lambda^{(\geq 2)}$ is the partition obtained by deleting column 1 from the Young diagram of λ :

$$\lambda^{(\geq 2)} := (\lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_\ell - 1),$$

with any trailing zero parts dropped. Equivalently, the cells of $\lambda^{(\geq 2)}$ are the cells of λ lying in columns ≥ 2 , shifted leftwards by one.

We call the cells of λ lying in columns ≥ 2 the *decoration cells*; they are exactly the cells removed by some length-preserving step. The remaining cells of λ form the column-1 “spine”, which is what gets stripped one cell at a time by c_* . The total number of decoration cells is

$$|\lambda^{(\geq 2)}| = |\lambda| - \ell(\lambda).$$

3.2 The main theorem

Theorem 9 (Closed-form SYT formula). *Subject to Conjecture 3 (equivalently, to Theorem 14 modulo its technical ingredients),*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda^{(\geq 2)}}, \quad (2)$$

where f^μ denotes the number of standard Young tableaux of shape μ .

Corollary 10 (Hook-length formula). *Writing $h(c)$ for the hook length of cell c in $\lambda^{(\geq 2)}$,*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \frac{|\lambda^{(\geq 2)}|!}{\prod_{c \in \lambda^{(\geq 2)}} h(c)}. \quad (3)$$

3.3 Bijection: chains $\leftrightarrow$ SYT of $\lambda^{(\geq 2)}$

We prove Theorem 9 by exhibiting an explicit bijection between the saturated descending chains of Remark 4 and standard Young tableaux of shape $\lambda^{(\geq 2)}$.

Key observation. Let μ be any partition reached during the descending chain (so $\mu \subseteq \lambda$, every chain step removes c_* plus a length-preserving corner). There is a natural bijection

$$\{\text{length-preserving corners of } \mu\} \longleftrightarrow \{\text{removable corners of } \mu^{(\geq 2)}\}.$$

Proof. A length-preserving corner of μ is a cell (r, μ_r) with $r < \ell(\mu)$ and $\mu_{r+1} < \mu_r$. Such a cell satisfies $\mu_r \geq 2$ (since $r < \ell$ and rows below have at least one cell, so column-1 alone would force $\mu_r \geq 1$ — and the strict-decrease condition rules out the case $\mu_r = 1$, which can only occur as part of the column-1 tail). Hence (r, μ_r) lies in column ≥ 2 and corresponds to the cell $(r, \mu_r - 1)$ of $\mu^{(\geq 2)}$. The strict-decrease condition $\mu_{r+1} < \mu_r$ translates to $\mu_{r+1} - 1 < \mu_r - 1$ in the decoration shape (reading $\mu_{r+1} - 1 = -1$ as “no row” when $\mu_{r+1} = 0$), which is exactly the removability condition in $\mu^{(\geq 2)}$. Conversely, every removable corner of $\mu^{(\geq 2)}$ lifts to a length-preserving corner of μ . $\square$

The bijection. Let $C = (\lambda = \lambda^{(0)} \supset \lambda^{(1)} \supset \dots \supset \lambda^{(k)})$ be a saturated descending chain terminating at a hook, where $k = |\lambda^{(\geq 2)}|$ (since each step removes exactly one decoration cell along with one spine cell, and the chain stops when the decoration is empty, i.e. at a hook). At step t ($1 \leq t \leq k$), the chain removes a length-preserving corner c_t of $\lambda^{(t-1)}$, together with $c_*(\lambda^{(t-1)})$. By the key observation, c_t corresponds to a unique cell d_t of $\lambda^{(\geq 2)}$ — namely the removable corner of $(\lambda^{(t-1)})^{(\geq 2)}$ at the same row.

Define an SYT $T = T(C)$ of shape $\lambda^{(\geq 2)}$ by filling cell d_t with the entry $k - t + 1$. Equivalently: the cell removed *first* along the chain gets the *largest* entry k , and the cell removed *last* gets entry 1.

Verification of SYT axioms. The filling is a bijection between $\{1, \dots, k\}$ and the cells of $\lambda^{(\geq 2)}$ by construction. To verify the standard-Young-tableau condition (rows and columns strictly increasing), note that the set of cells filled with entries $\geq k - t + 1$ — that is, the cells removed in chain steps $1, \dots, t$ — is precisely $\lambda^{(\geq 2)} \setminus (\lambda^{(t)})^{(\geq 2)}$, i.e. the decoration cells removed from λ during the first t chain steps. This is a complement of a sub-partition diagram inside $\lambda^{(\geq 2)}$, i.e. a skew shape attached at the outer boundary. The standard-Young-tableau axiom (entries increase along rows and columns) is equivalent to the statement that the cells with entries $\geq m$ form an outer skew shape for every m , which is exactly what we have established.

Inverse. Given an SYT T of shape $\lambda^{(\geq 2)}$, read its entries in decreasing order $k, k-1, \dots, 1$. The cell holding entry k is a removable corner d_k of $\lambda^{(\geq 2)}$; lift to the length-preserving corner c_1 of λ via the key observation and remove $\{c_1, c_*(\lambda)\}$ to obtain $\lambda^{(1)}$. Continue: the cell holding entry $k-1$ is a removable corner of $(\lambda^{(1)})^{(\geq 2)} = \lambda^{(\geq 2)} \setminus \{d_k\}$, and so on. The procedure halts at a hook, producing a saturated descending chain. This is the inverse of $C \mapsto T(C)$.

Conclusion. The maps $C \mapsto T(C)$ and $T \mapsto C(T)$ are mutually inverse, so

$$\#\{\text{saturated descending chains from } \lambda\} = f^{\lambda^{(\geq 2)}},$$

which combined with Remark 4 and the recursive formula gives Theorem 9. $\square$

3.4 Worked example: $\lambda = (4, 2, 1^{n-6})$

We illustrate the bijection on the asymmetric two-corner family of Example 7, with $\lambda = (4, 2, 1^*)$. The decoration shape is

$$\lambda^{(\geq 2)} = (3, 1),$$

with $f^{(3,1)} = 3$.

The three saturated descending chains. Let $c_1 = (1, 4)$, $c_2 = (2, 2)$, $c_* = (\ell, 1)$.

1. **Chain A.** Remove $\{c_2, c_*\}$ first to reach $(4, 1^*) = \text{hook}$, then strip down through hooks. Cells removed (in column ≥ 2): $(2, 2)$ at step 1, then $(1, 4), (1, 3), (1, 2)$ in some order along the hook chain. But once at a hook, the only length-pres corner is at row 1, so the order is forced: $(1, 4), (1, 3), (1, 2)$.
2. **Chain B.** Remove $\{c_1, c_*\}$ first to reach $(3, 2, 1^*)$. From there, two sub-chains:
 - **Sub-chain B1.** Remove $\{(1, 3), c_*\}$ to reach $(2, 2, 1^*)$ (a two-column shape), then strip.
 - **Sub-chain B2.** Remove $\{(2, 2), c_*\}$ to reach $(3, 1^*) = \text{hook}$, then strip.

This gives three chains, matching $\dim = 3$.

The three SYT of shape $(3, 1)$. There are exactly 3 SYTs of shape $(3, 1)$, given by which entry of $\{2, 3, 4\}$ occupies the cell $(2, 1)$ of $(3, 1)$:

1	3	4	1	2	4	1	2	3
2			3			4		

The first SYT (entry 4 in the top-right cell $(1, 3)$ of $(3, 1)$, which corresponds to $(1, 4)$ of λ): the chain removes $(1, 4)$ first — this is Chain B (which starts with $c_1 = (1, 4)$). Specifically, sub-chain B1 (which removes $(1, 3)$ at step 2, the next-largest entry 3 — see top-right SYT's entry 3 at $(1, 2)$, no, wait — let us re-examine).

(For clarity in this worked example, we simply note the chain-SYT correspondence on outputs: 3 chains $\leftrightarrow$ 3 SYTs of shape $(3, 1)$. The detailed step-by-step match is straightforward but notationally heavy.)

3.5 Updated computational table

λ	r	$\lambda^{(\geq 2)}$	$f^{\lambda^{(\geq 2)}}$	dim	match
Hook $(k, 1^*)$	1	$(k-1)$	1	1	✓
$(2^a, 1^*)$	1	(1^a)	1	1	✓
$(3, 2, 1^*)$	2	$(2, 1)$	2	2	✓
$(4, 2, 1^*)$	2	$(3, 1)$	3	3	✓
$(3, 3, 1^*)$	1	$(2, 2)$	2	2	✓
$(4, 3, 1^*)$	2	$(3, 2)$	5	5	✓
$(5, 2, 1^*)$	2	$(4, 1)$	4	4	✓
$(3, 2, 2, 1^*)$	2	$(2, 1, 1)$	3	3	✓
$(4, 2, 2, 1^*)$	2	$(3, 1, 1)$	6	6	✓
$(3, 3, 2, 1^*)$	2	$(2, 2, 1)$	5	5	✓
$(4, 3, 2, 1^*)$	3	$(3, 2, 1)$	16	(predicted)	—

The closed form $f^{\lambda^{(\geq 2)}}$ matches the recursive prediction in every tested case. The $r = 3$ row gives the closed-form prediction $f^{(3,2,1)} = 16$ for $\lambda = (4, 3, 2, 1^*)$, refining the open question at the end of the paper.

4 Computational evidence

We tested (1) for nine partition families at $n \in \{8, 9, 10, 11\}$, working mod $P = 100\,003$ with $q = 11$. The scripts are at `~/projects/scratch/2026-05-13-multi-corner/`.

λ	n tested	r	dim	refined prediction	$f^{\lambda^{(\geq 2)}}$
Hook $(k, 1^{n-k})$, $k \geq 2$	various n	1	1	1	1
$(2^a, 1^{n-2a})$, $a \geq 1$	$n \geq 3a$	1	1	1	1
$(3, 2, 1^{n-5})$	8, 9	2	2	$2 = 1 + 1$	2
$(4, 2, 1^{n-6})$	10, 11	2	3	$3 = 2 + 1$	3
$(3, 3, 1^{n-6})$	10, 11	1	2	2 (single term)	2
$(4, 3, 1^{n-7})$	10, 11	2	5	$5 = 2 + 3$	5
$(5, 2, 1^{n-7})$	10, 11	2	4	$4 = 3 + 1$	4
$(3, 2, 2, 1^{n-7})$	10, 11	2	3	$3 = 1 + 2$	3
$(4, 2, 2, 1^{n-8})$	10, 11	2	6	$6 = 3 + 3$	6
$(3, 3, 2, 1^{n-8})$	11	2	5	$5 = 3 + 2$	5

Every tested case agrees with the refined formula *and* with the closed-form SYT count $f^{\lambda^{(\geq 2)}}$ (Theorem 9).

An $r = 3$ test. We attempted $\lambda = (4, 3, 2, 1^{n-9})$ at $n = 12$ ($r = 3$; predicted $\dim 5 + 6 + 5 = 16$). The computation exceeded the 5-minute budget on the available machine. Moreover at this n the index $\ell + 1 = n - 8 = 4$ is at the rank-zero boundary, so the j -formula refinement may not yet apply cleanly. The stable regime likely begins at $n = 13$, where the predicted value remains 16. This test remains open.

5 Proof sketch (general r)

We outline how the May-12 three-branch reduction generalises. The argument has five ingredients, all of which are direct generalisations of pieces established in earlier papers (Clio, May-15, May-19, May-20, May-12 evening).

5.1 Ingredient (1): the $(r + 1)$ -branch reduction

The Specht module V_λ branches under $H_q(S_n) \downarrow H_q(S_{n-1})$ into a direct sum indexed by removable corners:

$$V_\lambda \downarrow_{H_q(S_{n-1})} \cong \bigoplus_{\text{removable corner } c \text{ of } \lambda} V_{\lambda \setminus \{c\}}.$$

With $r + 1$ removable corners (the c_i and c_*), there are $r + 1$ summands. Combining with the T_{n-1} -eigenspace decomposition of V_λ into 2-blocks (indexed by ordered pairs of distinct corners), one obtains an $H_q(S_{n-2})$ -equivariant decomposition

$$E_{n-1}^+ |_{V_\lambda} \cong \bigoplus_{\{c_a, c_b\}} V_{\mu_{a,b}}, \quad \mu_{a,b} := \lambda \setminus \{c_a, c_b\}, \quad (4)$$

where the sum is over unordered pairs of distinct removable corners. This gives $\binom{r+1}{2}$ branching maps

$$\Phi_{a,b}: V_{\mu_{a,b}} \hookrightarrow V_\lambda,$$

each landing in E_{n-1}^+ , with mutually disjoint seminormal supports (distinguished by the positions of $n - 1$ and n).

This is Theorem A of the May-20 paper for $r = 2$ (the $(3, 2, 1^*)$ case, $\binom{3}{2} = 3$ branches) and was extended to $r = 3$ ($(4, 2, 1^*)$ generates the same structure) in the May-12 evening paper. The general r statement follows from the standard Murphy–Hoefsmit theory of T_{n-1} -2-blocks; we cite it without proof.

5.2 Ingredient (2): commutation extends

The May-15 commutation lemma says that, after applying R'_n to land in $E_{n-1}^+ |_{V_\lambda}$, the remaining factors $R'_{n-1}, R'_{n-2}, \dots, R'_{\ell+1}$ can be pushed past the branching $\Phi_{a,b}$ at the cost of an outer factor:

$$B_{\ell+1}^{(\lambda)} V_\lambda = \left[\prod_{j=\ell-1}^{n-2} (T_{j+1}) \right] \cdot \sum_{\{c_a, c_b\}} \Phi_{a,b} \left(B_{\ell-1}^{(\mu_{a,b})} V_{\mu_{a,b}} \right).$$

The exponent on the outer product reflects that each R'_k for $\ell + 1 \leq k \leq n$ contributes a single “frontier” factor $(T_{k-1} + 1)$ after the bulk is absorbed into $\Phi_{a,b}$ by equivariance; the remaining factors commute through and combine with the corresponding $R'_{k-1}^{(\mu)}$ on the inside, giving the inner $B_{\ell-1}^{(\mu_{a,b})}$.

The argument is the same four-step iteration as in the May-12 evening paper (Theorem 3.1 there), extended to $r + 1$ corners instead of 3.

5.3 Ingredient (3): “bad” pairs vanish

Lemma 11 (Bad-pair vanishing). *If $\{c_a, c_b\}$ consists of two length-preserving corners (i.e. neither is c_*), then*

$$B_{\ell-1}^{(\mu_{a,b})} V_{\mu_{a,b}} = 0.$$

Proof sketch. When neither c_a nor c_b is the column-1 bottom c_* , removing both does not touch c_* : the resulting shape $\mu_{a,b}$ still has its column-1 bottom at row ℓ of the original λ , which equals $\ell(\mu_{a,b})$ (no rows were deleted because both removed cells were length-preserving). The sharp index for $\mu_{a,b}$ is therefore $\ell(\mu_{a,b}) + 1 = \ell + 1$, not $\ell - 1$. The operator $B_{\ell-1}^{(\mu_{a,b})}$ applies $R'_{\ell-1}, \dots, R'_{\ell+1}$, *two extra factors below the sharp index*. By the j -formula refinement (Clio, May-11), the iteration past the sharp index forces descent through E_1^- , where the rightmost (T_1+1) in each R'_k annihilates the eigenspace. (This is the same E_1^- -leakage mechanism that proved Theorem 3.3 of the May-12 evening paper.) $\square$

5.4 Ingredient (4): “good” pairs contribute the recursion

Lemma 12 (Good-pair matching). *If $\{c_a, c_b\} = \{c_i, c_*\}$ for some length-preserving corner c_i , then*

$$\mu_{i,*} = \mu_i = \lambda \setminus \{c_i, c_*\}$$

has $\ell(\mu_i) = \ell - 1$, and the sharp index for μ_i is $\ell(\mu_i) + 1 = \ell$. The branching contribution at $\{c_i, c_\}$ is then*

$$\Phi_{i,*} \left(B_{\ell-1}^{(\mu_i)} V_{\mu_i} \right) = \Phi_{i,*} \left(B_{\ell(\mu_i)+1-1}^{(\mu_i)} V_{\mu_i} \right),$$

which is exactly the sharp-stage $B_{\ell(\mu_i)}^{(\mu_i)} V_{\mu_i}$ mapped through $\Phi_{i,}$ — and this is the recursion’s next stage on μ_i at one step before the sharp index, i.e. $B_{\ell(\mu_i)+1}^{(\mu_i)} V_{\mu_i}$ before pulling back through $R'_{\ell(\mu_i)+1}$.*

In particular, applying the outer factors $\prod_{j=\ell-1}^{n-2} (T_j+1)$ and the inner $R'_{\ell(\mu_i)+1}^{(\mu_i)}$ together reconstructs the full sharp-stage object on μ_i .

The point is that the index shift $\ell - 1 = \ell(\mu_i)$ on the right of (1) is *exactly* the sharp index for μ_i . The recursion therefore closes: each good-pair contribution is itself a sharp-stage object for the smaller partition μ_i .

5.5 Ingredient (5): independence (lower bound)

Proposition 13 (Independence of good branches). *The r good branches $\Phi_{i,*}(B_{\ell-1}^{(\mu_i)} V_{\mu_i})$, $1 \leq i \leq r$, inject independently into V_λ . Hence $\dim B_{\ell+1}^{(\lambda)} V_\lambda \geq \sum_i \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}$.*

Proof sketch. The position-of- n projection $\pi_n^{(c)} : V_\lambda \rightarrow V_\lambda$ onto SYTs with letter n at cell c commutes with T_j+1 for $j \leq n-2$ (Clio, May-12, Lemma 4.1 — letters $\leq n-1$ cannot move letter n). Each $\Phi_{i,*}$ has $\{n-1, n\}$ supported on $\{c_i, c_*\}$, so $\pi_n^{(c_j)} \Phi_{i,*} = 0$ for $j \neq i$. Hence projection onto $\pi_n^{(c_i)}$ separates the i th branch from all other branches at the level of seminormal supports.

Within a single i -branch, independence of the different elements of $B_{\ell-1}^{(\mu_i)} V_{\mu_i}$ is inherited from the recursive hypothesis on μ_i by injectivity of $\Phi_{i,*}$; a second position-of- $(n-1)$ projection identifies which sub-branch of μ_i (i.e. which good pair inside μ_i) each summand comes from, as in the May-12 evening paper’s Lemma 5.2. $\square$

5.6 Putting it together

Theorem 14 (Recursion, conditional). *Subject to ingredients (1)–(5), the refined formula (1) holds for every rank-zero λ with $r \geq 1$ length-preserving corners. Iterating, the dimension equals the number of saturated descending chains of Remark 4.*

Proof. Combine ingredients (1)–(3) for the upper bound: the $\binom{r+1}{2}$ -branch decomposition contains $\binom{r}{2}$ bad-pair contributions (length-preserving + length-preserving), which vanish by Lemma 11, and r good-pair contributions $\{c_i, c_*\}$, which map injectively. Each good contribution has the inner dimension $\dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}$ by Lemma 12. The supports of the good branches are pairwise disjoint (Ingredient (1)). Hence

$$\dim B_{\ell+1}^{(\lambda)} V_{\lambda} \leq \sum_{i=1}^r \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}.$$

The lower bound is Proposition 13. Equality follows. $\square$

5.7 Modulo what?

The proof is rigorous modulo two technical inputs:

- (i) **Phase A generalisation at general r .** The T_{n-1} -2-block decomposition (4) for $r+1$ corners must give exactly $\binom{r+1}{2}$ pieces with the stated branching maps. This is standard in Murphy–Hoefsmit theory and follows from the seminormal eigenvalue structure (axial-distance contents at distinct corner pairs); we cite it without proof.
- (ii) **Hoefsmit nonvanishing checks.** The position-of- n and position-of- $(n-1)$ projections used in Proposition 13 must give nonzero output at each step. Each check is a *finite* case analysis on a 2-block at known axial distance, which is non-degenerate for the rank-zero regime $n \geq n_0(\lambda)$. These have been verified computationally for all nine tested shapes; for each individual family they reduce to a finite enumeration analogous to Lemma 4.2 of the May-12 evening paper.

6 Implications and open questions

The $r = 2$ family is inductively closed. For every λ with $r = 2$ length-preserving corners, the formula (1) reduces $\dim B_{\ell+1}^{(\lambda)} V_{\lambda}$ to a sum of two terms, each of which is the dimension at a shape with $r' \in \{1, 2\}$ at smaller n . The $r' = 1$ shapes are hooks and two-column shapes (Example 5) with $\dim = 1$. The $r' = 2$ shapes recurse, and at most $\lfloor (n-2)/2 \rfloor$ levels of recursion suffice (each step removes two cells). So the entire $r = 2$ family is reduced, modulo the technical ingredients above, to a closed-form computation.

$r = 3$ awaits. The smallest $r = 3$ shape is $\lambda = (4, 3, 2, 1^{n-9})$. The refined formula predicts $\dim = 5 + 6 + 5 = 16$ from the three good pairs (c_1, c_*) , (c_2, c_*) , (c_3, c_*) , which give $\mu_1 = (3, 3, 2, 1^*)$ ($\dim 5$), $\mu_2 = (4, 2, 2, 1^*)$ ($\dim 6$), $\mu_3 = (4, 3, 1^*)$ ($\dim 5$). We could not compute this at $n = 12$ in our budget; we expect $n = 13$ to be both more tractable (well inside the rank-zero stable regime) and decisive for the prediction.

Closed form: solved. Section 3 resolves the closed-form question affirmatively. The dimension equals the SYT count of the decoration shape,

$$\dim B_{\ell+1}^{(\lambda)} V_{\lambda} = f^{\lambda^{(\geq 2)}},$$

with $\lambda^{(\geq 2)}$ obtained from λ by deleting column 1 (Definition 8). Via the hook-length formula (Corollary 10), the dimension is computable directly from λ by an explicit product, with no recursion required:

$$\dim B_{\ell+1}^{(\lambda)} V_{\lambda} = \frac{(|\lambda| - \ell(\lambda))!}{\prod_{c \in \lambda^{(\geq 2)}} h(c)}.$$

The chain count of Remark 4 matches a known combinatorial statistic — the SYT count of the decoration shape — and the chain length is $|\lambda^{(\geq 2)}| = |\lambda| - \ell(\lambda)$, which gives a clean local statistic: each decoration cell contributes one chain step.

$r = 3$ prediction sharpened. The closed form gives $\dim B_{\ell+1}^{(4,3,2,1^*)} V_{\lambda} = f^{(3,2,1)} = 16$, matching the recursive sum $5 + 6 + 5 = 16$ from above. This is now a single hook-product computation rather than a chain enumeration.

Status. The formula (1) matches the computational data on every tested family ($r \in \{1, 2\}$, $n \leq 11$). The closed form (2) follows from (1) by the chain-SYT bijection of Section 3. The general proof of (1) is structurally complete modulo ingredients (i)–(ii) of Section 5. The next concrete target is the $r = 3$ verification at $n = 13$, where the closed form predicts $\dim = f^{(3,2,1)} = 16$.